

Assignment: Mini Search Engine using Vector Search

Objective:

Build a Google-like semantic search application using Streamlit and Pinecone where users can upload PDF documents and query them using natural language.

Problem Statement

Develop a Mini Search Engine that allows users to upload at least five PDF documents. The system should extract text, split it into chunks, generate embeddings, store them in Pinecone, and retrieve relevant content based on user queries.

Functional Requirements

1. Create a Streamlit web application.
2. Allow users to upload at least 5 PDF files.
3. Extract text from uploaded PDFs.
4. Split extracted text into smaller chunks.
5. Generate embeddings for each chunk.
6. Store embeddings and metadata in Pinecone.
7. Accept a user search query.
8. Convert the query into an embedding.
9. Retrieve Top-K relevant chunks from Pinecone.
10. Display document name, similarity score, and retrieved text.

Suggested User Interface

Search Engine

[Upload PDFs]
[Search Box]
[Search Button]

Results:

- PDF name
- Similarity score
- Retrieved paragraph

Deliverables

- Source code (Github link)
- Dataset (5 PDFs)
- Deployed Streamlit App

Learning Outcomes

- Understand embeddings
- Use Pinecone vector database
- Implement semantic search
- Understand retrieval systems